

Safe Learning-Based Control Using Model Predictive Control and Gaussian Processes

by

Munachimso Nwosu

Queen's University

Department of Electrical and Computer Engineering

Master of Engineering

Under the guidance of

Dr. Melissa Greeff

and

Adam Hall

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Lay Summary

This research project explores a method to improve the future performance of a dynamic system using past system data. Navigating from one point to another while avoiding obstacles and changes in its environment and payload can be challenging, as these changes can lead to poor tracking performance and instability. A Model Predictive controller (MPC) is a control technique which works by repeatedly predicting the system's behavior over a finite horizon or time frame and picking the best action. Gaussian Process Model Predictive Control (GPMPC) is a learning-based control method for overcoming modelling errors. The error compensation is achieved by using data and a Gaussian Process (GP) model to make a reliable prediction. This model is then used inside an MPC. The learned GP error dynamics used inside the MPC help compensate for modeling errors, which provides a safer and more performative controller for UAV navigation. In this project, a Python-based GPMPC was developed for a 2D quadrotor. The decision-making of the MPC and the GP's learning and estimation will be studied and utilized to improve tracking performance, this was then evaluated in simulation on sinusoidal trajectories.

Introduction

Background

The utilization of aerial robots in search-and-rescue missions, especially in the face of natural disasters, demands autonomous operation in challenging and dynamic conditions. Complexities such as winds, various obstacles, active environmental changes, proximity to humans, and unknown mass distribution of a payload necessitate advanced control strategies that can correct modeling errors. Traditional control theory

proves stability and guarantees reliable motion under certain assumptions. In learning-based control, we use machine learning inside classical control frameworks to reduce the impact of these conservative assumptions, resulting in better performance while maintaining the guarantees provided by classical control. Classical control guarantees safety but is limited to a small part of the world, this makes it conservative in its performance. While a purely data-driven approach may enable complex high-performance control of unknown systems, most current approaches do not provide formal stability or safety guarantees. Combining both approaches may provide better performance by modeling more of the environment and can also provide strong safety guarantees, as seen in figure 1.

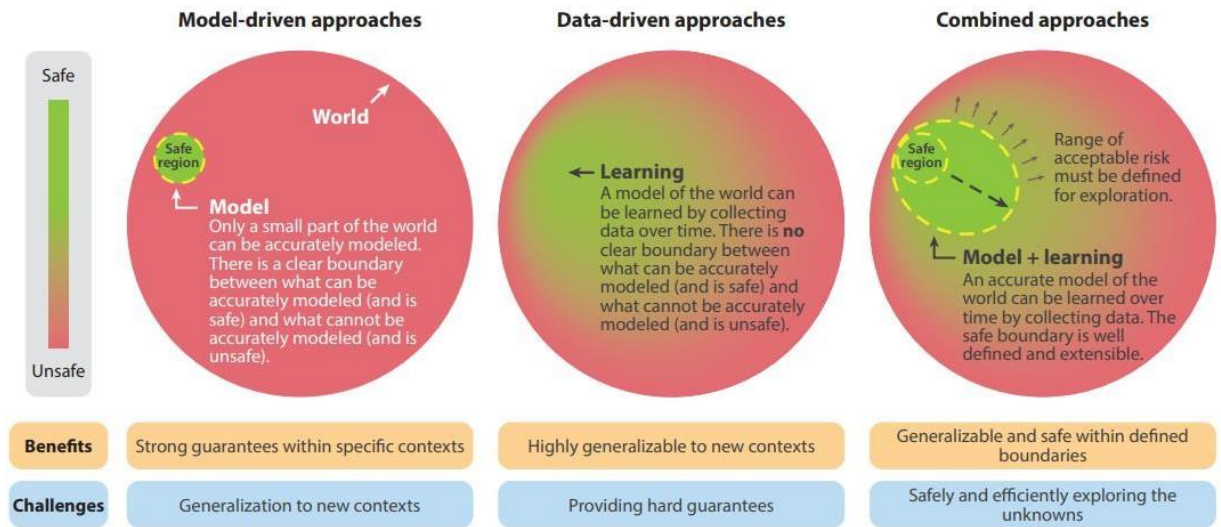


FIGURE 1. [14] “SAFE LEARNING IN ROBOTICS: FROM LEARNING-BASED CONTROL TO SAFE REINFORCEMENT LEARNING”

This research focuses on implementing a cutting-edge learning-based control approach, specifically GPMPC on a multi-rotor UAV for search-and-rescue operations. The project acknowledges uncertainties in the robot's dynamics, such as unknown mass

distribution during payload operations and poorly characterized operating environments in extreme weather conditions. The proposed MPC algorithm leverages expressive learning models to enable closed-loop control with safety guarantees.

Objectives and Scope

The project's primary objective is to develop a learning-based control algorithm for a quadrotor whose dynamics are uncertain. Specifically, we implement and investigate an approach known as GPMPc. The focus will be on utilizing state error information acquired from both the actual states and predicted states, along with an integration of a GP. The implementation of the project is limited to a simulated quadrotor in the X and Z dimensions, horizontally and vertically. The aim of the project is to control a quadrotor's movement through a given space by using MPC and using a GP to improve the system's trajectory over time. This project does not cover real-world quadrotor implementation. Still, it serves as a proof-of-concept to highlight the effectiveness of combining MPC with GP models for dynamic robust control in a simulated environment.

Scientific Summary

In this study of accurately modeling dynamics, using the combination of GP and MPC, the objective was to improve the decision-making of the system. By updating a prior model using the power of probabilistic modeling and real-time optimization control, the system was able to compensate for errors in its tracking. Through simulating the system, states and input controls were monitored in comparison to a reference trajectory to view the system's performance. These

results contribute valuable insights into the adaptability and robustness of GPMPC in a controlled, simulated environment.

Methodology

Theory

GPMPC emerges at the intersection of two powerful concepts in machine learning and control theory: GPs and MPC. GPs operate as a probabilistic model by capturing uncertainties that could exist in a system's dynamics. MPC determines the control inputs over a defined finite horizon by solving an optimization problem at each time step. This combination addresses challenges posed by uncertain and complex systems where traditional models and control techniques may fall short.

At the core of GPMPC lies the Gaussian process, a non-parametric model capable of characterizing distributions over functions. By employing GPs to represent system dynamics, GPMPC departs from the rigidity of traditional deterministic models. Instead, it embraces uncertainty, providing a versatile framework for dealing with incomplete knowledge of the underlying uncertain dynamics. The GP's mean function and covariance function kernel facilitate predicting system states and associated uncertainties.

Two-Dimensional Plane Attitude Model

The states, x position, z position, $\dot{x}$ velocity, $\dot{z}$ velocity, θ , and angular velocity are represented by the variables, $x, z, \dot{x}, \dot{z}, \theta, \dot{\theta}$. The states of the quadrotors are controlled by the input vector, $\mathbf{u}$. The pitch and thrust control, θ_{cmd} and T_{cmd} are optimized to manipulate the position, velocities, and pitch of the rotor. τ_θ and κ_θ are constants determined experimentally, the attitude dynamics are from [5].

$$\mathbf{u} = [\theta_{cmd} \quad T_{cmd}]^T.$$

EQUATION 1. CONTROL INPUTS

$$\mathbf{x} = [x \quad z \quad \dot{x} \quad \dot{z} \quad \theta \quad \dot{\theta}]^T.$$

EQUATION 2. STATES

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{z} \\ T_{cmd} \cdot \sin\theta \\ T_{cmd} \cdot \cos\theta \\ \dot{\theta} \\ 1/\tau_\theta (\kappa_\theta \theta_{cmd} - \theta) \end{bmatrix}.$$

EQUATION 3. 2D ATTITUDE MODEL

Error Dynamics

The model in equation 4 is in continuous time. To discretize it a fourth order Runge-Kutta was used. The system's dynamics is represented in discrete time by the following equation:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{g}(\mathbf{x}_k, \mathbf{u}_k),$$

EQUATION 4. SYSTEM DYNAMIC

where $\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k)$ is an inaccurate prior model representing the dynamics of the system. The term $\mathbf{g}(\mathbf{x}_k, \mathbf{u}_k)$ represents the unmodeled dynamics. Using Gaussian process regression on error dynamics, we model $\mathbf{g}(\mathbf{x}_k, \mathbf{u}_k)$. Both are combined and a solution for the next time step is optimized.

Root Mean Square Error

To compare the performance of MPC and GPMPC we will compute the RMSE error. The root mean square between actual trajectory and the reference trajectory x and z positions were calculated to see how much better one system is compared to the other. The RMSE included the x and z position values taken from first till the last data point, N .

$$RMSE = \sqrt{\frac{\sum_{i=0}^N (x_i - x_{ref})^2 + (z_i - z_{ref})^2}{N + 1}}$$

EQUATION 5. RMSE CALCULATED FROM X AND Z POSITIONS

Cost Function

$$\begin{aligned} & \underset{\{u_k\}_{k=0}^{N-1}}{\text{Minimize:}} J_N(x, u) \\ & \text{Subject to: } x_{k+1} = f(x_k, u_k) \\ & \quad x(0) = x_0 \\ & \quad x_{min} \leq x_k \leq x_{max}, \quad \forall k \in [0, N] \\ & \quad u_{min} \leq u_k \leq u_{max}, \quad \forall k \in [0, N - 1] \end{aligned}$$

EQUATION 6. OPTIMAL PROBLEM

MPC aims to solve an optimal control problem where we would like to minimize the cost function by manipulating the control inputs, $\mathbf{u}$. This is subject to the dynamic constraint, $\mathbf{x}_{k+1}$ and the control action $f(\mathbf{x}_k, \mathbf{u}_k)$, the initial state value, and the minimum and maximum state and control action across the horizon (number of steps to look ahead) for all prediction steps. This optimization problem is solved at each step by using the optimal control input, the first input in the horizon is then applied to the real system.

One of the significant strengths of GPMPC is its ability to handle uncertainty robustly.

Traditional MPC relies on precise system models, but GPMPC thrives in scenarios where the system model is uncertain or evolving. The probabilistic nature of GPs in GPMPC offers not only point estimates but also insights into the confidence of the predictions. This is especially advantageous in real-world applications where uncertainties are inherent and dynamic.

$$\ell(x, u) = \|x_u - x^r\|_Q^2 + \|u\|_R^2.$$

EQUATION 7. RUNNING COST

The running cost is a quadratic function defined by taking the second norm of the difference between the predicted and reference states x_u and x^r and the control action, u . Q and R are weighted matrices used to prioritize certain states or control over the other(s).

$$J_N(x, u) = \sum_{k=0}^{N-1} \ell(x_k, u_k).$$

EQUATION 8. COST FUNCTION

The purpose of the cost function is to capture the system's performance objectives. The MPC controller system's behavior is best optimized by reducing penalties for violating constraints and minimizing the difference between its prediction and reference across the prediction horizon. By taking a summation of the costs between the horizon, the MPC optimizes to pick the best control action for the current and future states.

GP Prediction

Squared exponential or Radial Basis Function (RBF) was used in GP to calculate the covariance between the set of data points, x and z . σ , the output scale controls the magnitude of the function values predicted, and the lengthscale, ℓ determines how quickly the correlation between the data points decreases, see [13].

$$k(x, z) = \sigma \exp\left(-\frac{(x-z)^2}{2\ell^2}\right), \quad x, z \in \mathbb{R}^{m+n}.$$

EQUATION 9. RADIAL BASIS FUNCTION

In addition to the RBF kernel, the Matern kernel, another popular choice of covariance function for GP, was also utilized to see how performance would be affected. The form of the function for

the values of ν equal to $5/2$ is provided in equation 10. ν is a positive parameter that controls the smoothness of the function, see [18]

$$k_M^{\frac{5}{2}}(x, z) = \left(1 + \frac{\sqrt{5}|x-z|}{\ell} + \frac{5|x-z|^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}|x-z|}{\ell}\right).$$

EQUATION 10. MATERN FUNCTION

By applying these kernel functions to all pairs of training points, the GP kernel matrix is structured, and the prediction equation is defined as:

$$\widetilde{g}(X^*) = k(X^*, X)[K(X, X) + \sigma_n^2 I]^{-1}Y.$$

EQUATION 11. PREDICTION FUNCTION

Where $k(X^*, X)$ is a vector, the matrix, and σ_n^2 is a noise variance multiplied by an identity matrix to account for uncertainty. Y is vector of stacked data points, the training targets.

$$K(X, X) = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_N) \\ \vdots & \ddots & \vdots \\ k(x_N, x_1) & \cdots & k(x_N, x_N) \end{bmatrix}.$$

EQUATION 12. COVARIANCE MATRIX

$$P(y|f(x)) = N(f(x), \sigma^2).$$

EQUATION 13. GAUSSIAN LIKELIHOOD

In GP Gaussian likelihood is a probabilistic model used to describe the uncertainty of the target data, y . N is a Gaussian distribution with a mean at $f(x)$ and a noise variance, σ^2 , used to represent uncertainty, the noise is needed to maximize the likelihood. A marginal log-likelihood(MLL) was used to quantify how well the model explains the observed data. MLL is the

logarithm of the likelihood of the observed training data given the model parameters and hyperparameters.

In the GP, a confidence region is an area in space where the predicted range of values is expected to fall with a high probability. The region is centered around the mean prediction of the uncertain model.

The GPMPC workflow unfolds through several key steps. In this project it's assumed that we have an accurate state estimate. The MPC is initially ran with the system dynamic in equation 4, a poor dynamic to simulate potential error. The control inputs are determined by solving an optimization problem that considers both the predicted states and associated uncertainties.

Improbability in error dynamics is then modeled using the GP, enabling the prediction of future states including their associated uncertainties over a prediction horizon. In the next run of the MPC, the learned dynamics is passed into the MPC. This flexible approach allows for adaptive control strategies, particularly beneficial when faced with evolving or uncertain system behavior.

Literature reviews

Review 1: Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning

The journal, [14], focuses on safe learning in robotics, specifically in the context of control and reinforcement learning(RL). The authors aim to provide a comprehensive overview of recent advances in using machine learning to achieve safe decision-making under uncertainties.

The review begins by highlighting the increasing complexity of real-world robotic applications and the uncertainties that arise from various sources, such as imperfect modeling, noisy sensor measurements, and unknown operating environments. The emphasis on the importance of safe

behavior in these applications, both during the learning process and in the optimized policy of the controller stated. The review then discusses the traditional approaches taken by the control theory and reinforcement learning communities to address the problem of decision-making under uncertainties. Traditionally, Control theory has relied on model-driven approaches, leveraging known dynamics models, and providing guarantees under known operating conditions. On the other hand, reinforcement learning has taken a data-driven approach, which allows for adaptability to new contexts but lacks formal guarantees.

The text discusses the challenges and approaches in reinforcement learning (RL) for sequential decision-making due to uncertainty. RL is highlighted as a framework that does not require prior knowledge of dynamics models, unlike traditional control methods. The initial randomized actions of its algorithm due to a lack of explicit assumptions and constraints can limit its applicability to safe control. The exploration-exploitation dilemma in RL during learning is identified as a venerable challenge to address this dilemma. Bayesian inference is proposed as a solution, by computing posterior distributions over states, probable dynamics, or total cost from observed data, they can be used as a guide for exploration.

The review acknowledges previous works that have focused on specific techniques or approaches without emphasizing the safety aspect as well as highlights the need for a unifying perspective on safe learning control, bridging the gap between the control and reinforcement learning communities. It concludes by emphasizing the importance of realistic physics-based benchmarks to facilitate fair comparisons between control and reinforcement learning approaches. It also highlights some open challenges in the field of robot learning such as dealing with nonlinearity, conflicting objectives, it suggests potential future research directions.

In the benchmark, three different approaches for reinforced learning were discussed. First, the text compares the performance of learning uncertain dynamics using a learning-based robust MPC with a GPMPC. The GPMPC stabilizes a quadrotor subject to state and input constraints and demonstrates improved performance compared to linear MPC, especially when the system operates with inaccurate prior models. The GPMPC accounts for model inaccuracies and satisfies constraints with a high percent confidence interval.

In conclusion, the literature review provides a thorough overview of recent developments in safe learning in robotics by concentrating on the integration of control theory and reinforcement learning. It highlights the importance of safe behavior and discusses traditional approaches; it proposes a combined approach that leverages control theory's and reinforcement learning's strengths. The review also identifies knowledge gaps and suggests the application of machine learning in future research.

Review 2: Learning safety in model-based Reinforcement Learning using MPC and Gaussian Processes

The objective of the research in [16] is to propose a learning method for safe MPC-based RL that estimates the safety set (i.e., the set of MPC parameters that yield a controller able to satisfy constraints) in a data-driven manner. The authors aim to enhance the learning process with a probabilistic safety mechanism using GP regression.

The proposed method consists of three components: a parametric MPC scheme, an episodic reinforced learning algorithm, and GP regressors. The parametric MPC scheme is employed as a model-based controller with limited knowledge of the system's dynamics. The RL algorithm adjusts the MPC parametrization to improve its performance. GP regressors are used to estimate

constraints on the MPC parameters directly from data, predicting the likelihood of safe and unsafe policies.

The proposed method in this paper does not require additional assumptions on the prediction model to maintain computational flexibility. The authors illustrate the effectiveness of their method through a numerical example involving the control of a quadrotor drone in a safety-critical environment.

While the proposed method addresses the challenge of safety in MPC-based RL, there are still knowledge gaps and potential future research directions to explore. One knowledge gap is the difficulty in obtaining certificates on safety and stability in the MPC-based RL paradigm. Future research could focus on developing techniques to provide rigorous guarantees of safety and stability in this context. Additionally, the proposed method relies on GP regression for estimating constraints, and further investigation could explore alternative regression techniques or combinations of multiple regression methods to improve accuracy and efficiency.

In conclusion, the paper attempts to present a method for learning safety in MPC-based Reinforced learning using GP regression. The proposed method integrates a parametric MPC method, an episodic RL algorithm, and GP regressors to estimate constraints and enforce safety in the learning process. The research contributes to the field of learning-based control by addressing the challenge of safety in MPC-based RL without requiring additional assumptions on the prediction model. However, there are still knowledge gaps and opportunities for future research to explore safety and stability guarantees and investigate alternative regression techniques.

This paper [34] provides its reader with an overview of the origins of MPC and the future it was heading towards. In this paper, multiple approaches to solving problems such as linearization, algorithms, and stable predictive controls are discussed and compared. The issues of instability and performance of a nonlinear system are addressed, which at that time held this system from being used in practical applications.

Review 4: Robust Model Predictive Control: A Survey

In this paper [33], an overview of the robustness in MPC is explored, techniques for handling, stability, and performance are covered. The applications of MPC offered is constrained to a linear time invariance system with constraints. When this paper was written, MPC was not readily applicable in an industrial setting. It states that the robust of an MPC was heavily dependent on the application. The paper also acknowledges the lack of computationally efficient algorithms.

Review 5: Stochastic Model Predictive Control: An Overview and Perspectives for Future Research

Stochastic MPC (SMPC) is discussed in this paper [31], it aims to account for stochastic uncertainties which isn't accounted for in Robust MPC. SMPC uses chance constraints, allowing for a specified probability level of constraint satisfaction, it seeks for a trade-off between the control objective and a probabilistic constraint satisfaction. An overview of different approaches and past developments are covered.

Review 6: Learning-Based Model Predictive Control: Toward Safe Learning in Control

This paper [32] highlights the tendency of choosing between performance optimization from constraint satisfaction and safety. In this paper, Learning the system dynamics, Learning the controller design, and MPC for safe learning are highlighted. It also discusses the tension

between stochastic methods from machine learning and deterministic or robust approaches in control. The review discusses related methods and viewpoints in the field of learning based and data control. The motivation behind data driven approach for control results from closed loop performance and specific parameterization of an MPC which often difficult analytically describe. While current methods have shown success, there is a need for development towards computationally elegant, efficient, and reliable control methods.

Review 7: Gaussian Processes for Regression

The paper [9] emphasizes the effectiveness of Gaussian processes in Bayesian analysis of neural networks, showcasing the exactness achieved through matrix operations and the estimation of hyperparameters from data. The Gaussian process formulation offers the advantage of exact combination of prior and noise models through matrix operations using maximum likelihood. The methods proposed are tested on the “MacKay's robot arm problem”, showing how well it performs in a real-world problem.

Review 8: Prediction at an Uncertain Input for Gaussian Processes and Relevance Vector Machines Application to Multiple-Step Ahead Time-Series Forecasting

The technical report [2] discusses non-linear models mapping input D -dimensional column vectors x to single-dimensional $f(x)$ outputs using GP or Relevance Vector Machines (RVM). The issue of making predictions, that is of evaluating $f(x^*)$ in the case where the input (x^*) is not deterministic, but rather random variables with known Gaussian distribution. The report highlights that for uncertain inputs, the predictive distribution of the regressors is not Gaussian. This issue is addressed by approximating the predictive distribution of GPs and RVMs by a Gaussian distribution, computing the exact mean and variance of the actual predictive distribution. This prediction is then fed back into the input vector, to make predictions.

Review 9: Gaussian Processes in Control: Performance Guarantees through Efficient Learning

In [30], he states that weaknesses in deep learning with artificial neural networks, attributes of deep-learning, and the need to collect a vast data sample can be avoided by employing GP models in control laws. safety of real-world systems is a relevant concern; the derivation of performance guarantees for deep learning-based control laws proves to be challenging. This is a problem because safety plays a vital role in the practical application of learning-based algorithms. GP provide a measurable uncertainty along with its prediction, making it a frequent application in learning-based control, recently. Challenges such as, tracking accuracy for a broad class of control laws, how training samples impact the certifiable control performance, how can we reduce the computational complexity are covered through chapter throughout the paper.

Review 10: Efficient Reinforcement Learning using Gaussian Processes

In his thesis paper [29], Deisenroth states that, RL involves an agent interacting with the environment, making decisions, perceiving sensory inputs, and receiving reward/penalty signals, a model of the world and the consequences of actions applied to the world are unknown in advance. RL faces challenges like exploration-exploitation trade-offs and requires many interactions in order to find a good strategy. The work combines concepts from Bayesian statistics, machine learning, optimal control, and system identification to address challenges in reinforcement learning, providing a comprehensive and principled approach.

Review 11: Gaussian Process Models for Robust Regression, Classification, and Reinforcement Learning

The thesis [1] emphasizes the importance of approximation techniques in Bayesian analysis, models for robust regression and applications in deep learning are discussed. Later chapters describe Laplace's method, Expectation Propagation, and Markov chain Monte Carlo sampling

for approximate inference in Gaussian process models. The practical applications of Gaussian process models in regression, robust regression, classification, and reinforcement learning are discussed, with references to experiments.

Review 12: Adaptive Receding Horizon Control for Constrained MIMO Systems

The text [28] introduces an adaptive control algorithm designed for open-loop stable, constrained, linear, multiple input multiple output (MIMO) systems. The adaptive controller consisted of an iterative algorithm that provides a set of candidate plant models based on input-output data and an MPC in an open-loop time invariant system, for output constraints for all the plants inside the model set. Adaptive control to a MIMO system is challenging to both input and output constraints, the algorithm proposed in the papers aims to address this issue. The integration of real-time set membership SM identification and receding horizon control aims to address the complexities associated with ensuring constraint satisfaction while updating the system model.

Review 13: Robust MPC with recursive model update

In [27] discusses an adaptive MPC algorithm designed for linear systems with model uncertainty and additive disturbances. Previous robust and stochastic approaches are suitable for unmodelled dynamics and rapidly changing environment but aren't ideal for slow changing or constant parametric uncertainties. The paper focuses on incorporating online learning and system identification within MPC and properties of MPC algorithm. By the end of the paper, a computationally tractable model predictive control algorithm with recursive parameter update was presented. An online optimization was solved as a linearly constrained quadratic program and was proven to be recursively sufficient.

Review 14: Adaptive MPC with Chance Constraints for FIR Systems

In [26] introduces an adaptive stochastic Model Predictive Control (MPC) strategy for stable linear time-invariant systems with additive uncertainties, handling probabilistic output constraints and hard input constraints. The approach, like some of the other papers discussed, considers MIMO systems, in this case represented by a finite impulse response model, it estimates parameters using a linear Recursive Least Squares (RLS) algorithm. Comparative results with an adaptive robust MPC algorithm from a previous work show that the proposed algorithm achieves lower cost in simulated cases with identical disturbance sequences.

Review 15: Adaptive MPC under Time Varying Uncertainty: Robust and Stochastic

The paper [25] addresses the challenge of utilizing adaptive Model Predictive Control (MPC) strategy for constrained uncertain systems. Specifically, it focuses on a linear system with bounded time-varying additive uncertainty and robust constraints. The proposed adaptive MPC framework is designed to handle both robust and probabilistic state constraints, along with hard input constraints, while ensuring recursive feasibility and stability. An SM Method-based model adaptation algorithm was used to estimate and update time-varying offset uncertainty, ensuring containment of the true uncertainty in the Feasible Parameter Set.

Review 16: Robust Discrete-Time Set-Based Adaptive Predictive Control for Nonlinear Systems

This research paper [24] proposes an adaptive Nonlinear Model Predictive Control (NMPC) design technique for constrained discrete-time nonlinear systems which are affected by both parametric uncertainties, time-varying disturbances, and a linear uncertain parameter. The approach provides a balance between adaptability and robust stability in the face of uncertainties

and disturbances. The proposed adaptive NMPC approach demonstrates good performance and stability for challenging nonlinear control problems.

Review 17: Learning Model Predictive Control for Iterative Tasks. A Data-Driven Control Framework

A learning Model Predictive Controller (LMPC), a good design for repetitive tasks, which learns from previous iterations to enhance closed-loop performance. [23] addresses repetitive tasks with unknown reference trajectories, such as in complex systems with nonlinear dynamics and aims to design an LMPC where the reference trajectory is unknown in advance. The experimental validation through application to an infinite horizon linear quadratic regulator with constraints (CLQR) showed that the proposed control logic converges to the optimal solution of the infinite optimal control problem.

Review 18: Adaptive MPC for Iterative Tasks

This paper [22] builds on the framework of the previous, like the preceding paper, it addresses the challenge of dealing with model uncertainties in MPC. It tackles uncertainties in constrained systems with parametric uncertainties, not stochastic. The Adaptive Learning Model Predictive Control (ALMPC) algorithm introduced, integrates robust MPC with (LMPC) to construct terminal constraints and cost iteratively. The algorithm incorporates an adaptive mechanism for handling uncertainties, which, over iterations, relaxes the bounds of constraints and reduces conservatism. This paper considers real-time model adaptation, while guaranteeing satisfaction of state-input constraints, this wasn't covered by the previous. In the result section, they proved if the uncertainty is learned at every iteration, the amount of constraint tightening is lowered for nominal states.

Review 19: Robust Adaptive Model Predictive Control for Guaranteed Fast and Accurate Stabilization in the Presence of Model Errors

This work [21] addresses the challenge of dealing with unknown or changing operational environments in applications like autonomous driving, assistive robotics, and unmanned aerial vehicles. The authors propose a combination of an ℓ_1 adaptive controller (ℓ_1 AC), a discrete time controller, with a robust model predictive controller (RMPC), to compute an optimal input to achieve fast and accurate stabilizations as well as high performance. “The ℓ_1 adaptive controller forces the system to behave close to a linear reference model despite the presence of parametric uncertainties.” The purpose of the (ℓ_1 AC) is to consistently reduce the modeling error at each time step, enabling accurate stabilization towards the origin. They demonstrate that compared to other models, RMPC- ℓ_1 demonstrates an accurate control of the system.

Review 20: Provably Safe and Robust Learning-Based Model Predictive Control

This paper [20] addresses the trade-off between robustness and performance in control theory by introducing a Learning-Based Model Predictive Control (LBMPC) framework. This control method aims to account for safety, stability, and optimality. LBMPC aims to show how performance is improved by using theory from robust MPC “... to check whether these same inputs keep a nominal model stable when it is subject to uncertainty.” Robust adaptive control combines the features of both robust control and adaptive control. It aims to provide stability and performance guarantees in the presence of uncertainties and parameter variations.

Review 21: Learning-Based Robust Model Predictive Control with State-Dependent Uncertainty

In this journal [8], the challenges of RMPC in the presence of state-dependent uncertainties are addressed. In Pin et al. (2009) a conservative based approach for nonlinear systems was studied, but in the case where a priori certainty bound isn't available, one would have to result to over-

approximations. Using a dynamic model with a nominal model and additive uncertainty, they achieve a less conservative result than a worst case bound for the uncertainty. By combining sets from GP with a linear MPC, they prove how the results are better when compared to other methods.

Review 22: Robust Constrained Learning-Based NMPC Enabling Reliable Mobile Robot Path Tracking

The algorithm proposed in [7] combines traditional NMPC with learning-based approaches to improve constraint satisfaction and adapt to uncertain environments. RC-MPC aims to identify inputs ensuring constraint satisfaction by considering a fixed estimate of model uncertainty but are generally conservative. The proposed algorithm, Robust Constrained Learning-based Nonlinear Model Predictive Control (RCLB-NMPC), combines a robot model with a dynamic nonparametric disturbance model. Disturbances represent deviations between the expected model and actual robot behavior, are learned over successive trials. These disturbances are models using a GP. This paper uses sigma-point transform to approximate the dynamics uncertainty propagation. This method used, are extended beyond the previous work in (Ostafew et al., 2015a), learning model uncertainty are used to compute robust state not real-world experience.

Review 23: Cautious Model Predictive Control using Gaussian Process Regression

The focus is on overcoming the challenges of accurately identifying models for nonlinear systems particularly when only approximate or nominal descriptions are initially available. GP is leveraged to learn nonlinear dynamics to improve prediction model and tighten constraints. A linearization to approximate the dynamics uncertainty propagation is used. The effectiveness of

the proposed methodology in [12] is demonstrated through simulation and a practical implementation.

Review 24: Data-Efficient Reinforcement Learning with Probabilistic Model Predictive Control

Reinforcement Learning is a well-established framework for autonomous policy. However, a major limitation is its data inefficiency, requiring impractically high interactions with the environment. While Gaussian processes (GPs) have been used for this purpose, existing algorithms like PILCO have limitations, including the need for a full planning horizon, reliance on parametrized policies, handling constraints, and unreliability near constraint boundaries, 4] introduces a novel RL algorithm that overcomes these limitations. The approach introduces probabilistic MPC with learned GP models, integrating uncertainty propagation through time. The MPC facilitates planning for relatively short horizons, thereby reducing computational burden and supporting applications with infinite horizons. This MPC framework enables immediate updates to the learned model, enhancing robustness against model inaccuracies.

Approach

The approach of the project is grounded in the principles of optimal control theory, with a focus on MPC for trajectory tracking as well as applied GP. The MPC algorithm is chosen for its ability to account for system dynamics and constraints. The objective is to show how employing GP in the modeled dynamics can account for uncertainties in the MPC's movement through a reference path. The software library, casADi was used for its auto differentiation features, allowed the program to efficiently compute gradients for the loss function, equation 8, and for its interface to many commonly used optimal solvers. The fourth-order Runge-Kutta method was utilized as the integration technique for the discretization and approximation of the continuous

problem of both the simulated dynamics and the prior model for the MPC. The prior model represents the original movement of the quadrotor before the error learning in the GP.

Within the optimization solver of the CasADi framework, instances of the variables needed for optimization, cost functions, vectors for states, and target states were passed. An Interior Point optimizer or IPOPT[17] was used as the choice of solver for this nonlinear function, at each time step. For all the experiments but the last, the horizon or number of look-ahead steps was set to 30 with a 0.05-time step with a simulation time of 10 seconds. For the last experiment, considering noise, it was run for 7 seconds. In the 10 second simulations, there was a total of 201 data points for each of the 6 states, this includes their initial position. Five different trajectories were tested in this experiment and are summarized in table 1. The references were a circular shape, a lemniscate, an increasing frequency, $\omega(t)$, in x, an increasing frequency in z, in z, and an increasing frequency in both axes simultaneously. For all the trajectories, the targets of the x and z movements were a cos and sin wave of different amplitudes and frequencies depending on the trajectory. At each time step the optimization solver would be called to get the next state. An error dynamic, $g(x_k, u_k)$ was added to the prior model to simulate the error added to create a slightly worse performance. The error dynamic model was identical to the prior model, except τ_θ and κ_θ were changed, for the prior model τ_θ and κ_θ were initially set to 0.1 and 1 but different values were used to study how the GP performance was reduced or enhanced by a poor model. The prior model τ_θ and κ_θ were set to 0.4 and 1.4 for most in the experiments.

$$\text{state error: } g(\hat{x}_k, u) = \hat{x}_{k+1} - f(\hat{x}_k, u_k)$$

Trajectories	X Target	Z Target
Oval	$0.02\cos(4 * t)$	$0.09\sin(4 * t)$
Lemniscate	$0.02\cos(4 * t) + \sin(2 * t)$	$0.09\sin(4 * t)$
Increasing frequency in X	$0.02\cos(4 * t * \omega(t))$	$0.09\sin(4 * t)$
Increasing frequency in Z	$0.02\cos(4 * t)$	$0.09\sin(4 * t * \omega(t))$
Increasing frequency in X and Z	$0.02\cos(4 * t * \omega(t))$	$0.09\sin(4 * t * \omega(t))$

TABLE 1 REFERENCE TRAJECTORIES THAT WERE TESTED.

The input control, the states, and the state error, the difference between the output state and prior model states at each step were then stored and passed to the GP training program. The lengthscales and output scale of the GP were then optimized to fit the collected training data by minimizing the MLL. Only a GP for the angular velocity was considered, as that is where the dynamics uncertainties are(i.e., τ_θ and κ_θ) . An Adam optimizer[11] with a learning rate of 0.1 and a marginal log-likelihood was used to observe data and was used to calculate the loss. To get the observed prediction, testing data is fed through the likelihood, using the calculated prediction a confidence region could be achieved.

Adding the learned parameters from the Gaussian process to the prior MPC model was accomplished by loading the hyperparameter; lengthscale and outputscale from the GPyTorch to a GP squared exponential posterior mean prediction function in CasADi. Using a kernel function and squared exponential covariance function from the safe-control-gym[3] repository, on GitHub, CasADi was able to make accurate predictions of the angular velocity error. These predictions were returned to the MPC and were integrated to account for error.

Milestones and Deliverables

Initially, to get a solid understanding of CasADi and MPC, a one-dimensional model was developed for simulation in MATLAB, this was then implemented in Python. The goal was to get it to drive from the 0 position in the x coordinate to a final position by fixing the yaw and only manipulating the pitch. After this was completed, a similar task was repeated for a 2-dimensional model, moving in the z and x coordinates. Next was to build on my understanding of regression. A two-independent variable and one-dependent variable model GP was developed, and the predictions and visualization of the output helped lay the foundation for the subsequent part of the project. The next part of the project involved developing the 2-dimensional model, passing the training and testing data, and viewing how well my model performed. The final part of the project involved loading the hyperparameters from the GP back to the MPC in CasADi and ensuring that the trajectory of the quadrotor behaves like it's expected to. Using functions available from "safe-control-gym"[3], modifications were made to the specifications of my program. The learned corrections were passed back to the prior model to get a more accurate motion.

Evaluation of Results

τ_θ and κ_θ were modified to magnify the learning power of the GP. The state weight matrices, Q , greatly affected the quadrotor states at each step. The θ and $\dot{\theta}$ weights were set to 0.01, the x and z velocity weights were set to 0.1, the weight of z was set to 80 and the x to 100. The low horizon value led to a faster computational time but poor performance for the system, which is to be expected due to low prediction capability, it has limited knowledge of the future which results in a non-optimal solution. In my application, a horizon of 30 steps performed well, a longer horizon required a longer time to run and achieve the same performance.

The MPC and GPMPC were ran on these trajectories. Each GP was trained for that specific trajectory. Running the MPC and GPMPC on different trajectories, the consistent results show that the GP is able to learn properly and increase the performance of an MPC as shown in table 2 and figure 3 to 17.

Next, the number of data points were increased, to study how they affect performance. In theory, increasing the number of data points should always improve the performance of the GP, but from the observed result, this assumption isn't true. For the 200 data point, training with above 40 percent negatively affected the GPMPC, this is likely because the GP is able to capture most of the data needed to account for model uncertainties with relatively little data. Above 40 percent, the system might not be learning much due to saturation. Training more data requires memory and CPU/GPU, in this case there would be no benefit to training with more data points.

The result of training using GP displayed that adding Gaussian noise with a mean of zero and magnitude of 1 to the training data yielded better results. It's possible that by introducing randomness, overfitting is avoided. Without added noise, the covariance matrix could become ill-conditioned, leading to poor optimization and prediction performance. This also affected the lengthscales of each of the trained states and inputs.

After GP's training ended the predicted error was added back to the prior model. The GPMPC states when compared to the MPC states and its reference trajectory showed positive results. The GP had successfully predicted and accounted for uncertainty with little difference between the error dynamics and learned error dynamics.

Results

A series of experiments were explored to understand how changing certain variable would affect the MPC tracking and GP's learning. These variables included, simulating different trajectories, adjusting the number of training, and testing data, tuning the model's dynamic , as well as the kernel used in the GP. Each time a variable was changed in each experiment, the GP was retrained, and new data points were used. During theses simulations, root mean squared error was calculated and utilized to capture how much better or worse each run was. For all experiment except the third, τ_θ and κ_θ were held at 0.4 and 1.4. For all experiment but the fourth, RBF kernel was used in the GP. In all experiment but the final, the quadrotor started from a point which was extremely close to the reference an area on the reference trajectory. Starting from the center or at point away from each of the trajectories led to issues where the GP was unable to learn properly, due to high uncertainty and from not being trained properly. For each of the experiments performed, the best RMSE values are bolded to highlight best performance.

Testing Different Trajectory

In the first experiment, 5 trajectories were tested, RMSE between the MPC and GPMPC positions and the reference positions were calculated. As seen in table 2, comparing the values of the MPC and GPMPC RMSE, it's evident that the GPMPC increases the tracking performance of the quadrotor, regardless of the trajectory, the RMSE of the GPMPC were all lower. Thirty percent of the total data acquired from the states and error difference were used for training the GPMPC. The RMSE of the lemniscate was recorded to be higher than the others. This could be due to the trajectory have a more rapid changes in direction at certain points.

Trajectory	MPC RMSE(m)	GPMP RMSE(m)	Percent error decrease (%)
Oval	0.0074533	0.0069062	7.34
Lemniscate	0.0354288	0.0281328	20.6
Increasing frequency in X	0.0074609	0.0067159	9.99
Increasing frequency in Z	0.0081493	0.0075901	6.86
Increasing frequency in X and Z	0.0090527	0.0084038	7.17

TABLE 2 MPC AND GPMP RMSE COMPARISON FOR DIFFERENT TRAJECTORIES

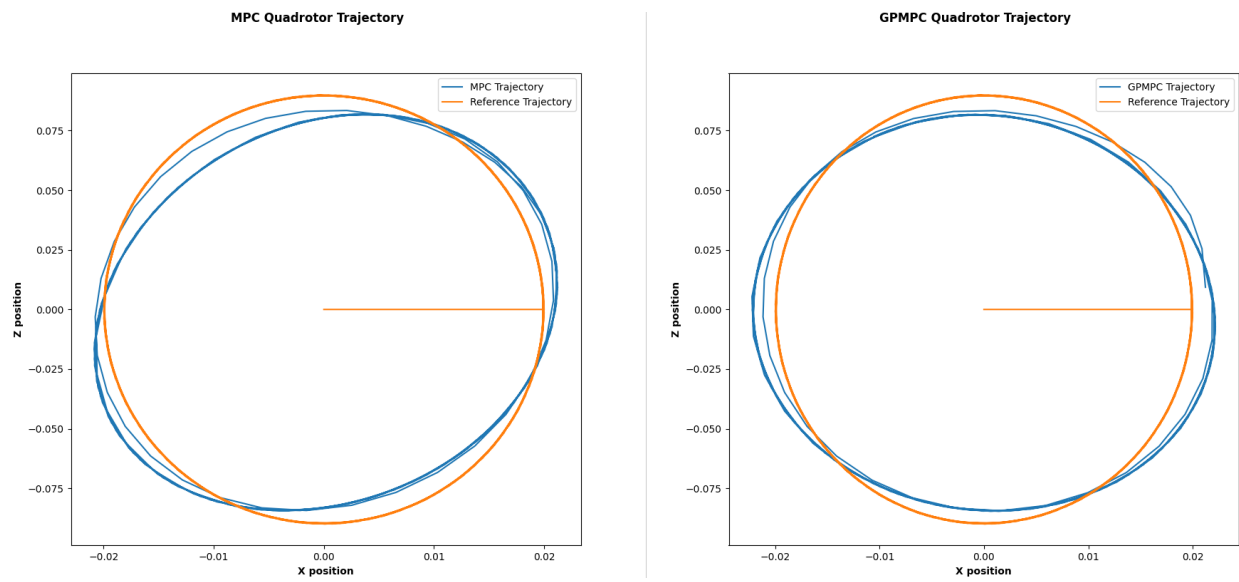


FIGURE 2 MPC VS GPMP OVAL TRAJECTORY

MPC

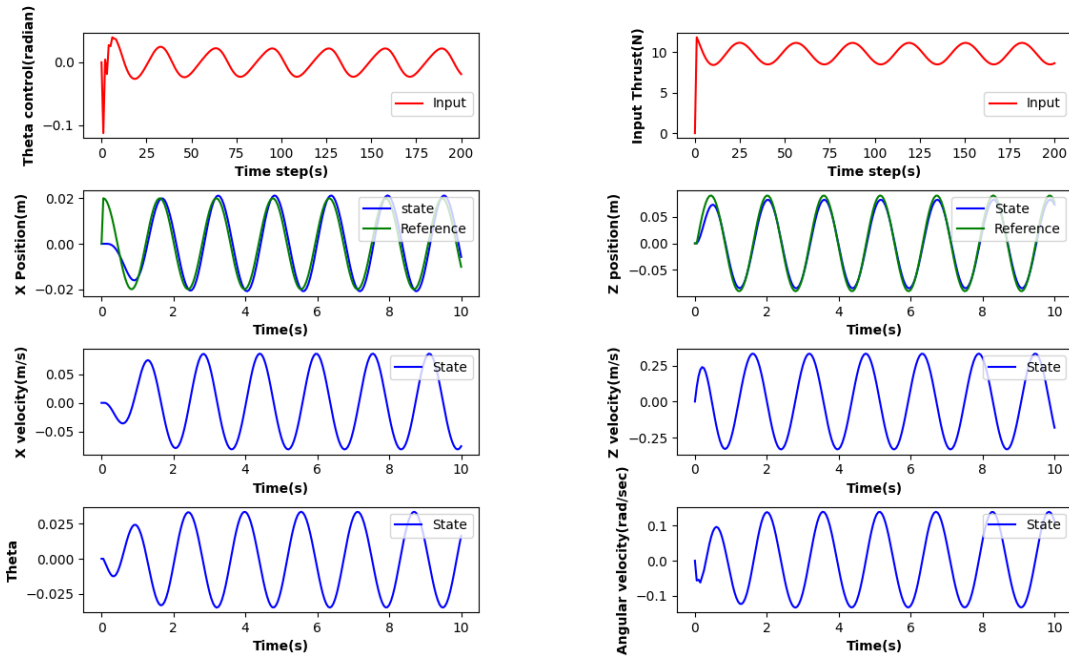


FIGURE 3 MPC STATES AND INPUTS FOR THE OVAL TRAJECTORY

GPMPC

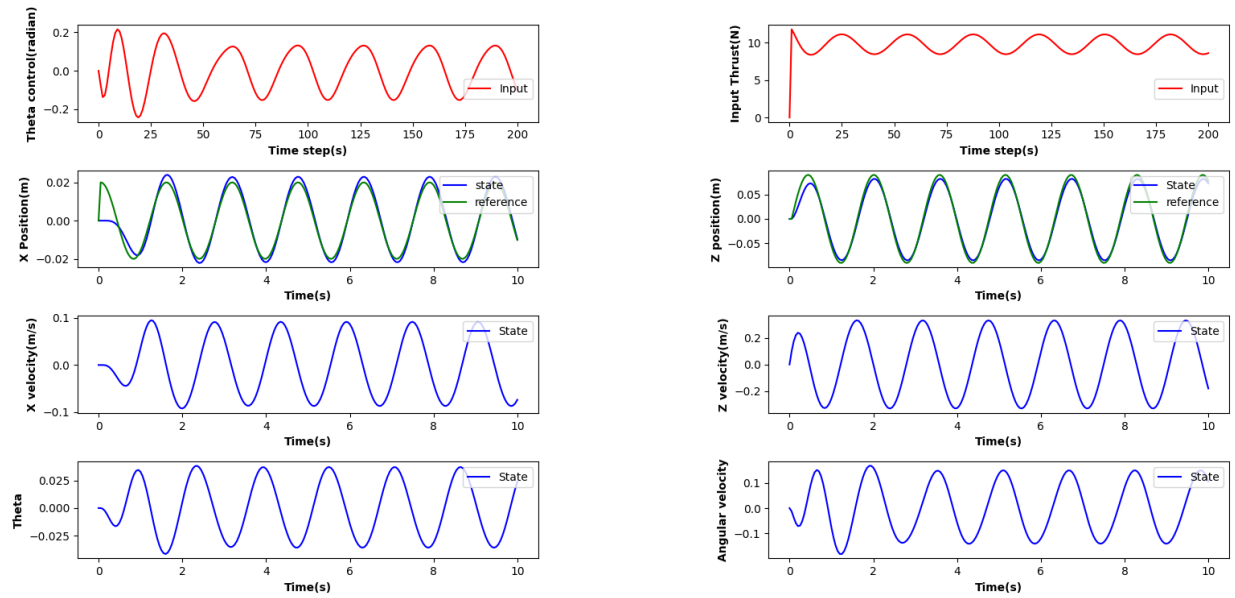


FIGURE 4 GPMPC STATES AND INPUTS FOR THE OVAL TRAJECTORY

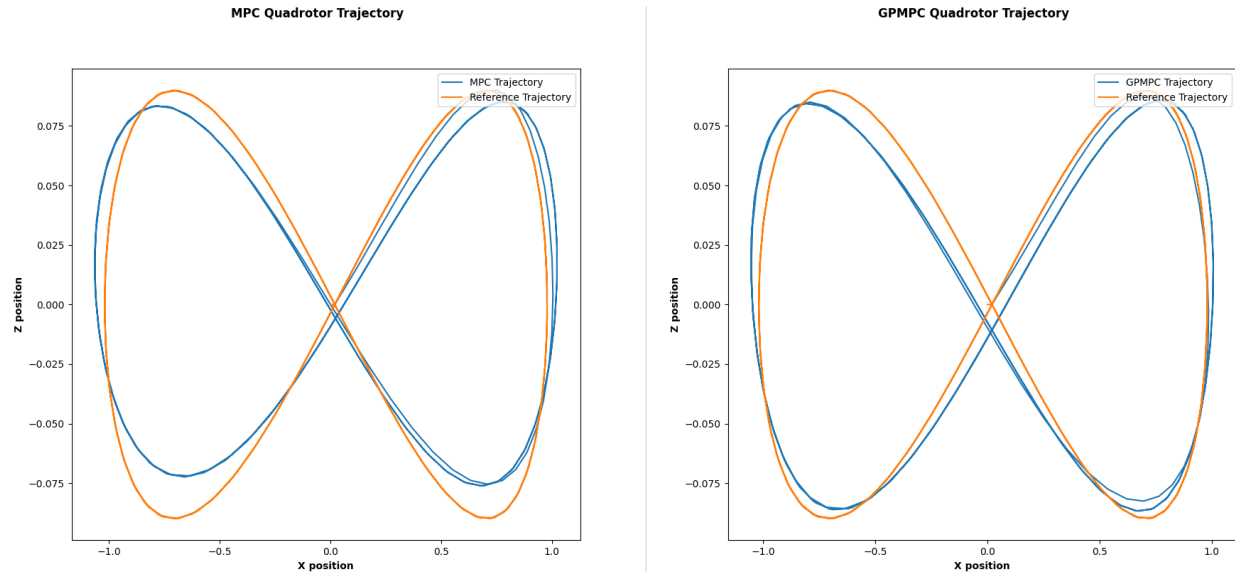


FIGURE 5 MPC VS GPMPC LEMNISCATE TRAJECTORY

MPC

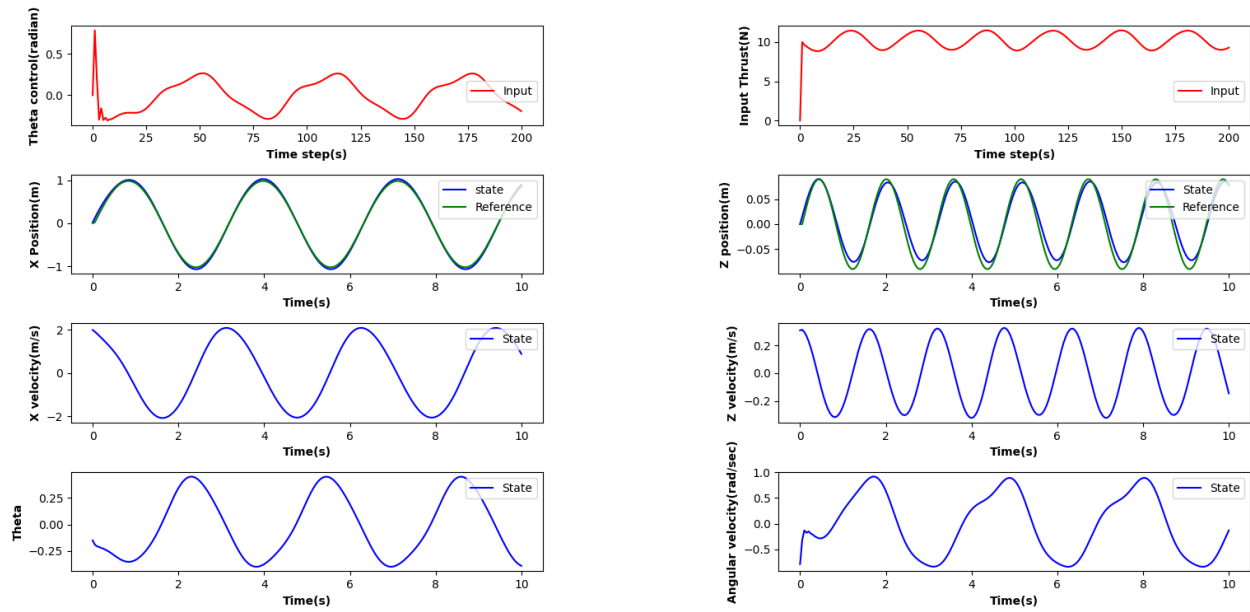


FIGURE 6 MPC STATES AND INPUTS ON FOR THE LEMNISCATE TRAJECTORY

GPMP

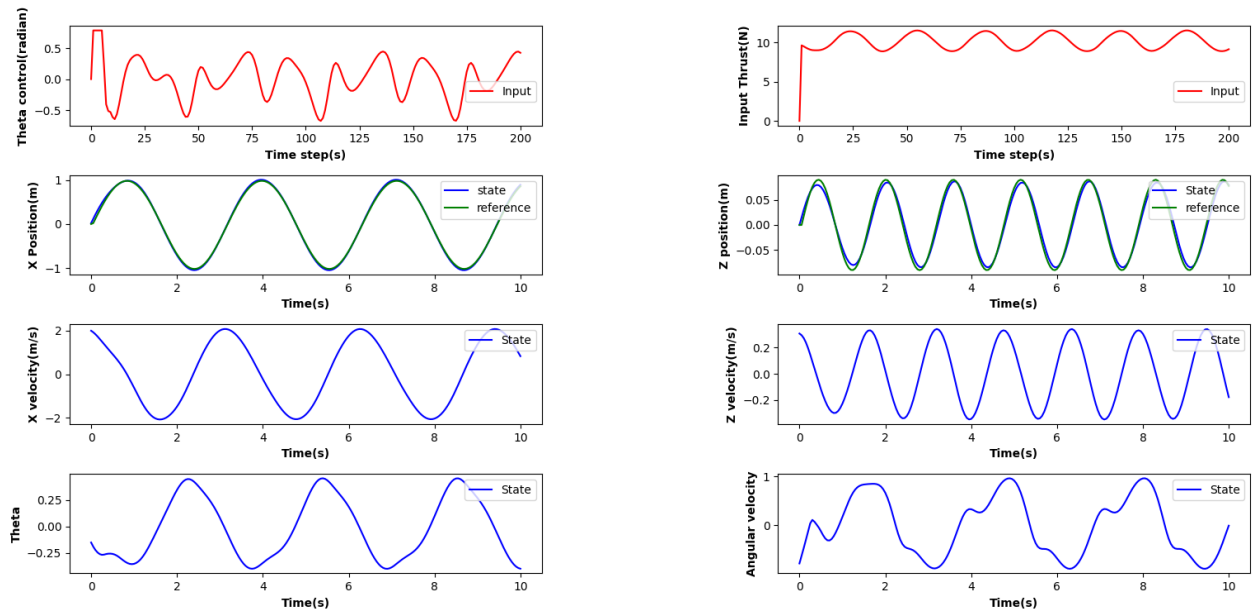


FIGURE 7 GPMP STATES AND INPUTS ON FOR A LEMNISCATE TRAJECTORY

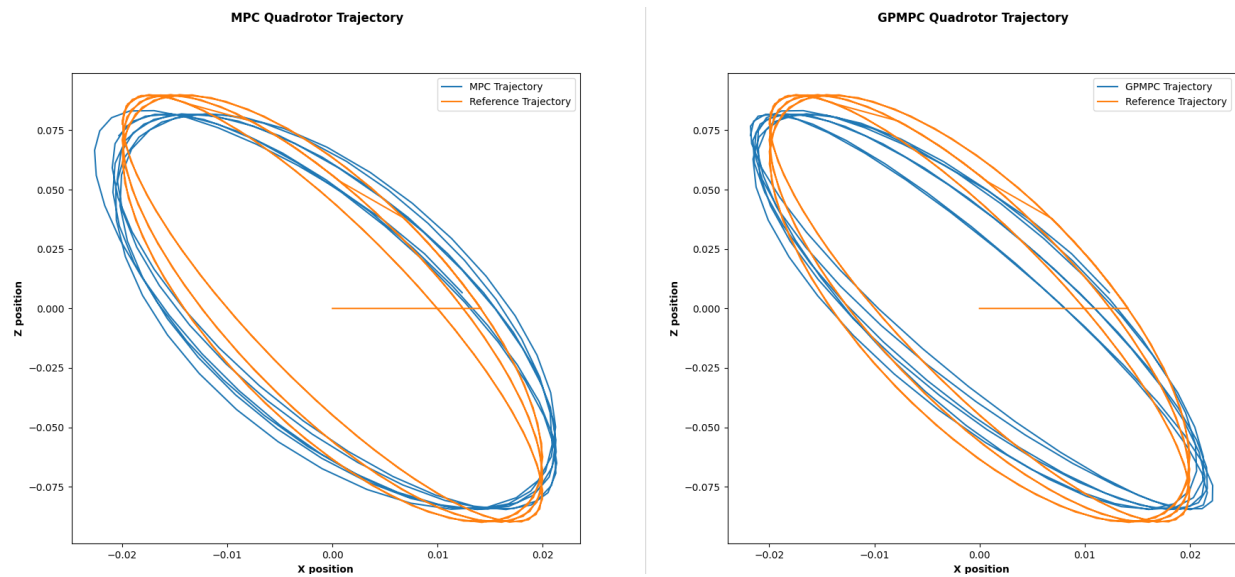


FIGURE 8 MPC VS GPMP OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X TARGET

MPC

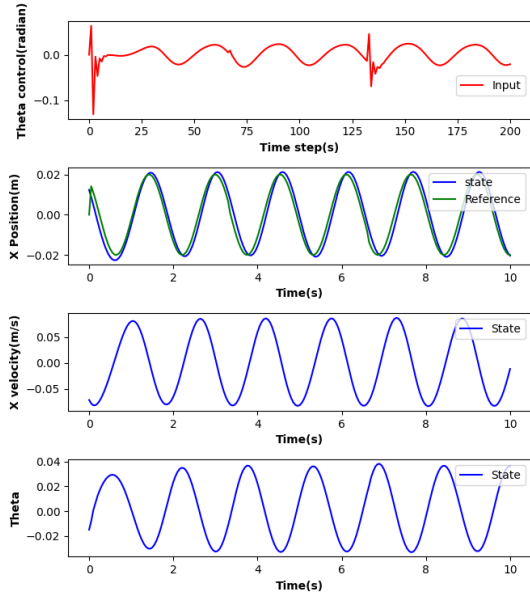
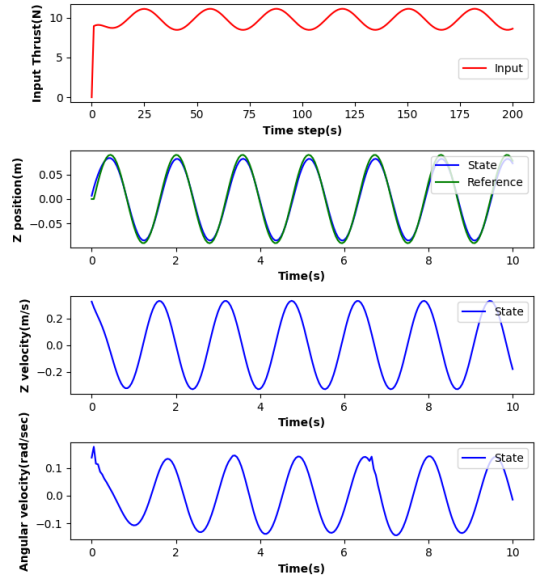


FIGURE 9 MPC STATES AND INPUTS ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X TARGET



GPMPC

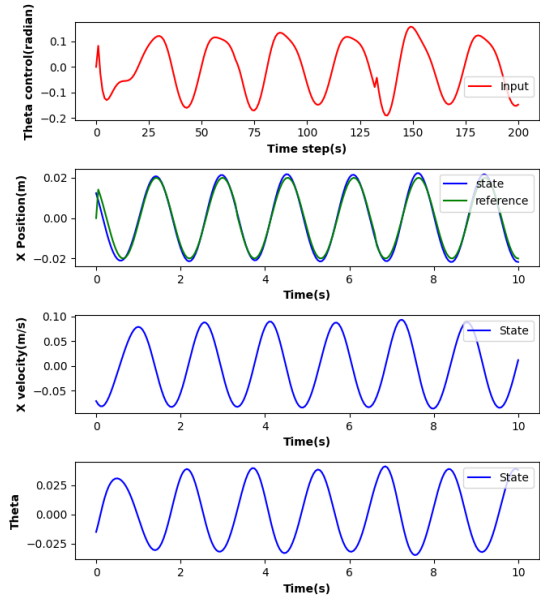
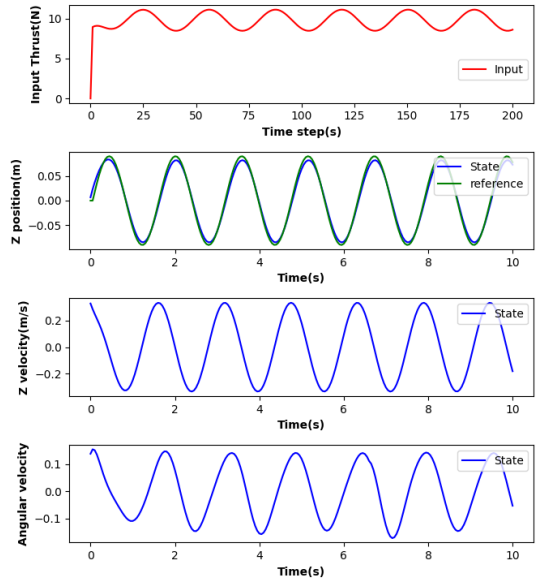


FIGURE 10 GPMPC STATES AND INPUTS ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X TARGET



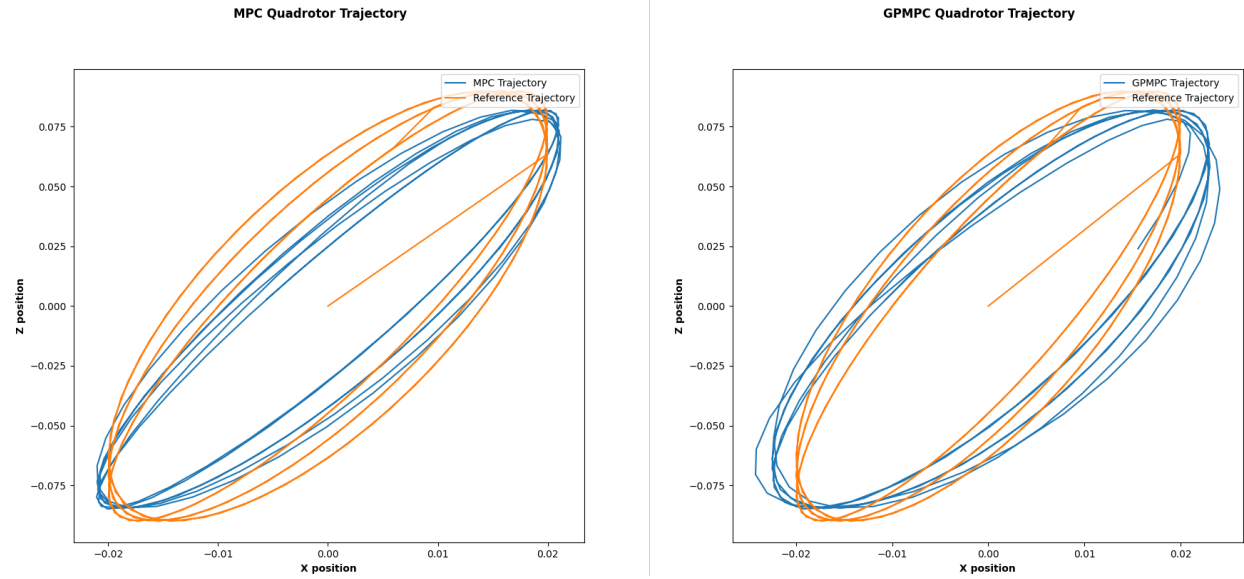


FIGURE 11 MPC VS GPMPC OVAL TRAJECTORY WITH INCREASING FREQUENCY OF Z TARGET

MPC

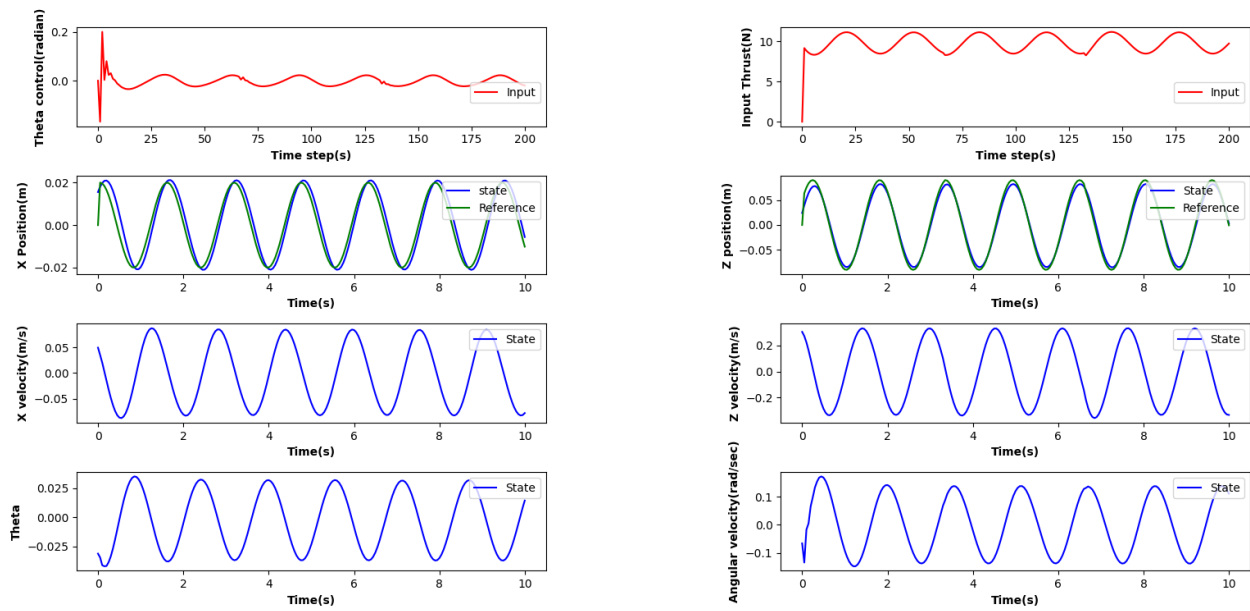


FIGURE 12 MPC STATES AND INPUTS ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF Z TARGET

GPMP

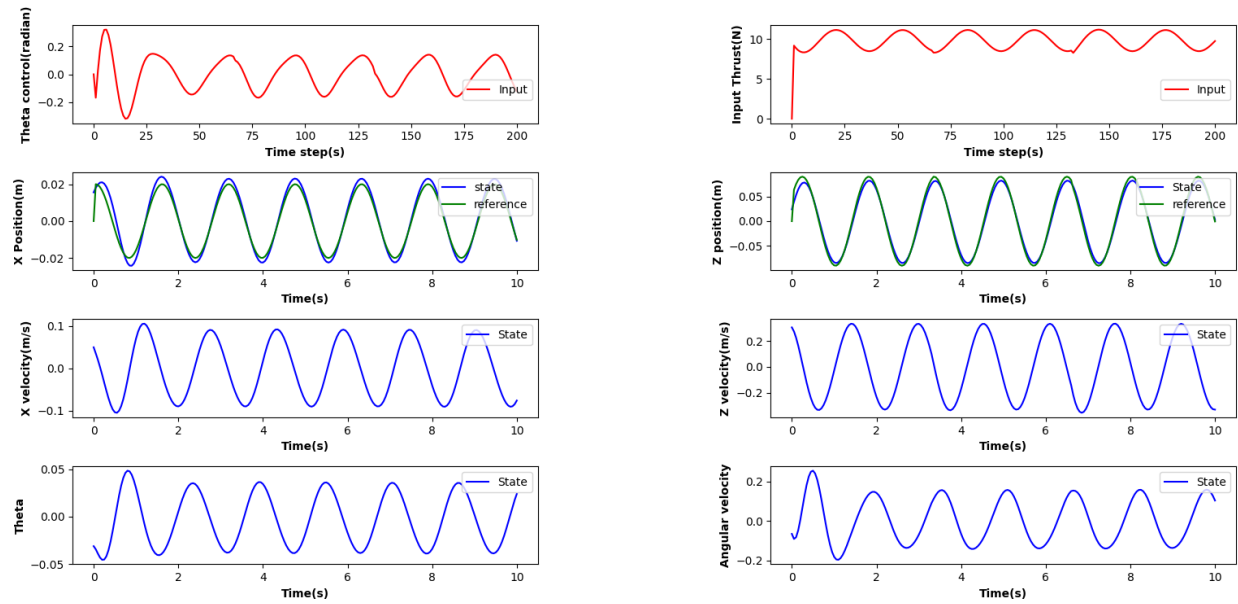


FIGURE 13 GPMP States and Inputs ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF Z TARGET

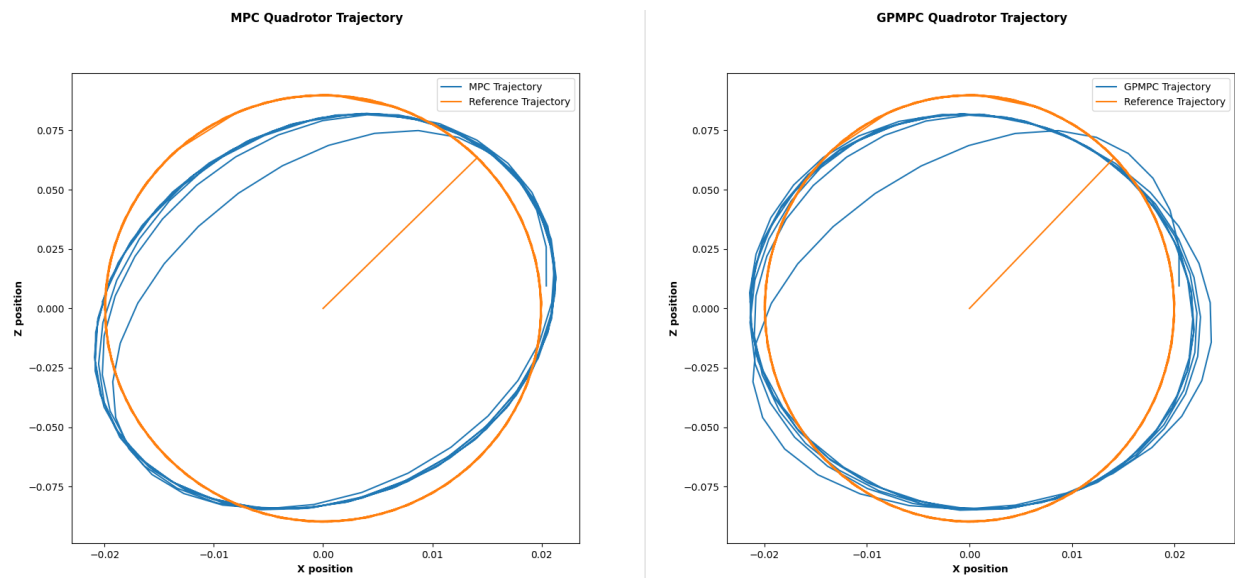


FIGURE 14 MPC VS GPMP OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X AND Z TARGET

MPC

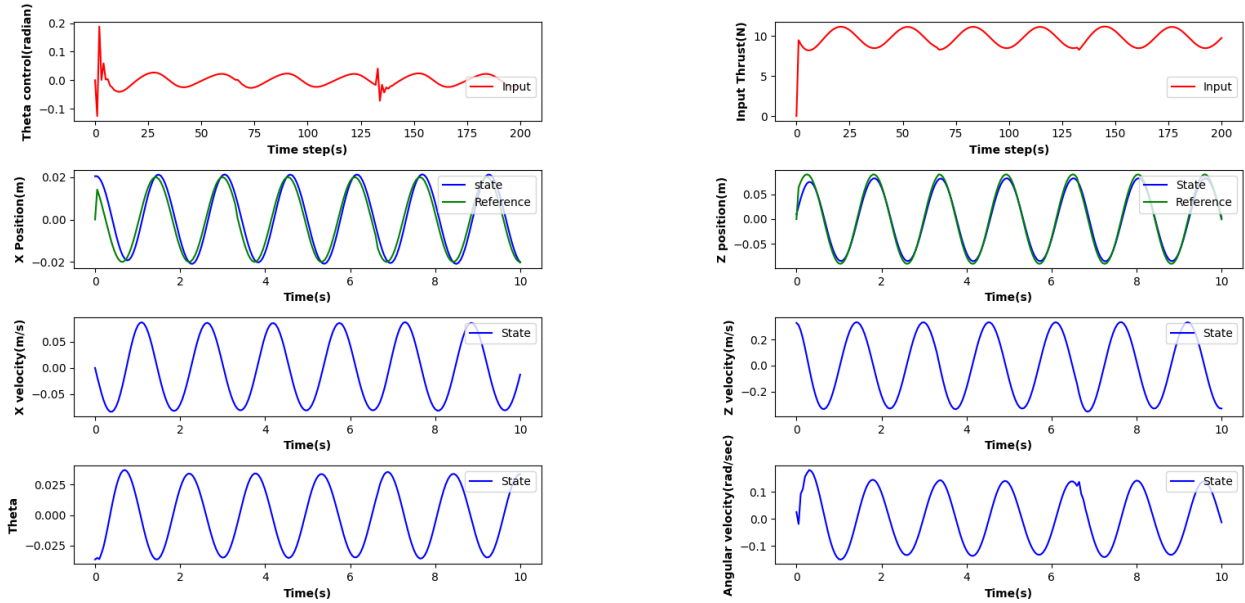


FIGURE 15 MPC STATES AND INPUTS ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X AND Z TARGET

GPMPC

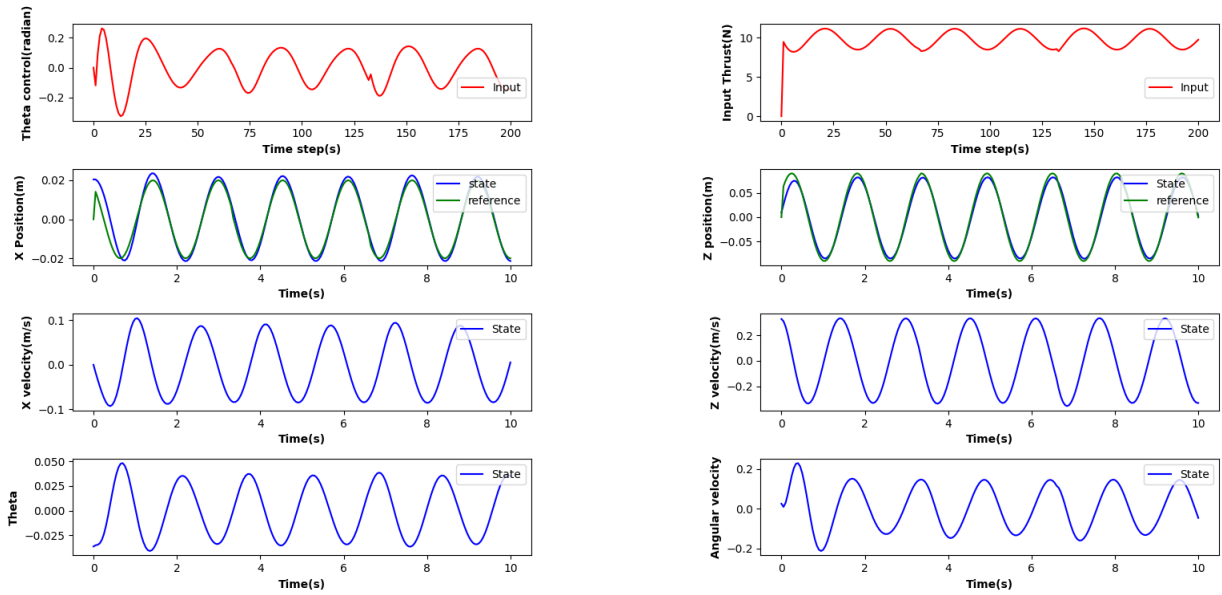


FIGURE 16 GPMPC STATES AND INPUTS ON FOR AN OVAL TRAJECTORY WITH INCREASING FREQUENCY OF X AND Z TARGET

Training with Multiple sizes of Data Points

The next part of this section explores on how different sizes of training data affects the MPC.

The oval trajectory with an Increasing frequency in x target was chosen, it had the lowest RMSE among the other GPMPC trajectories. From the results in table 3, it's clear that up to a certain

point, increasing the training dataset does not boost the performance of the GPMPC. An explanation could be that the GPMPC has collected and trained enough data from a certain region, increasing the number of data becomes redundant and only adds to noise.

Percent of Training point	GPMPC RMSE(m)
20%	0.0067172
30%	0.0067159
40%	0.0067152
60%	0.0067184
80%	0.0067235

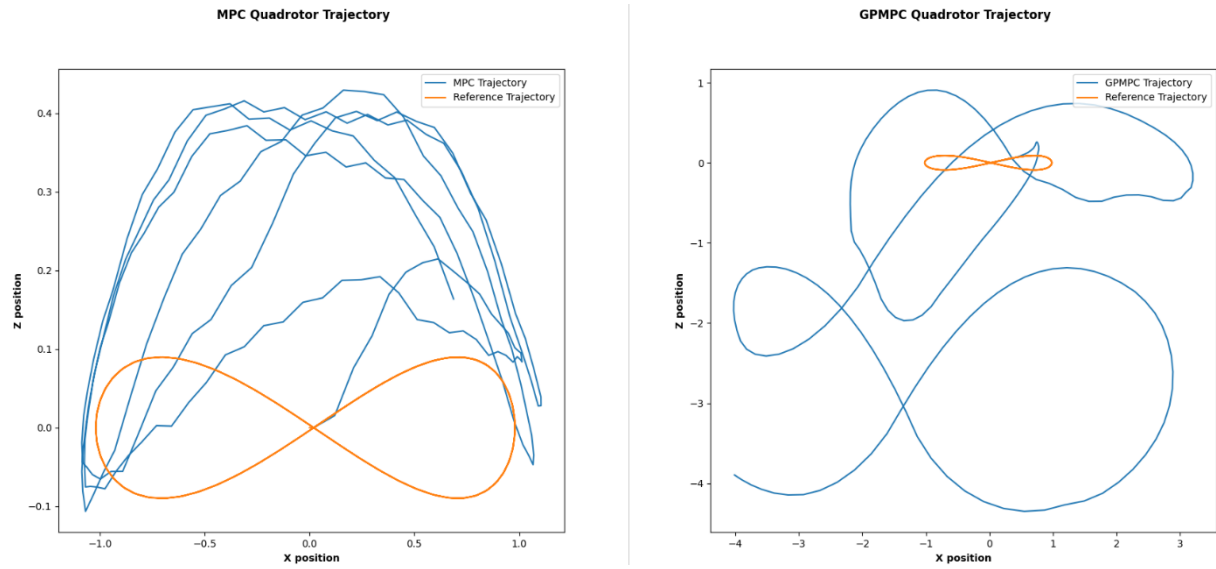
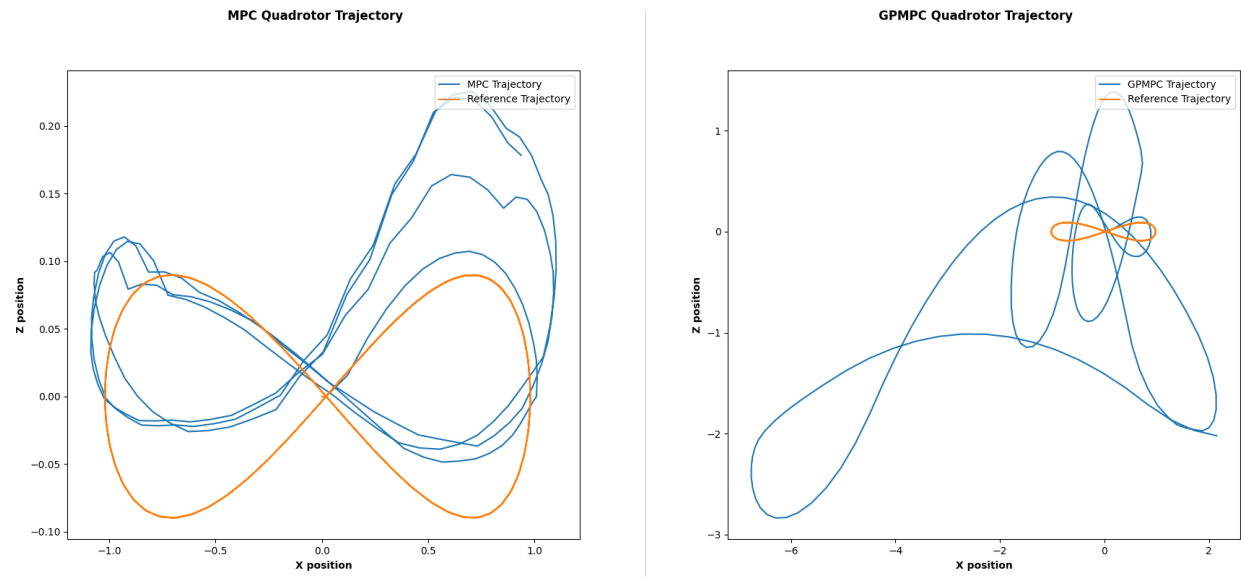
TABLE 3 DIFFERENT AMOUNTS OF TRAINING DATA ARE EXPLORED

Working with Different Dynamics

In this section the values of κ_θ are changed from 0.4 view how it affects both MPC and GPMPC. As observed in the table 4, the GPMPC dynamics is more sensitive to lower values of κ_θ , the frequency of the dynamics is low. Between 0.6 and 1 the performance on the GPMPC degrades and becomes unreliable. As seen from the plot below, for $\kappa_\theta = 0.1$ most of the region around the lemniscate aren't explore, this led to the GP performing poorly due to not having training data around the reference leading to unstable behavior. Another possible reason this is the case could be due to be GP optimization algorithm, when the GP is poorly trained, it can introduce many local minima into the optimization, leading to poor optimization of the GPMPC problem. Surprisingly, giving κ_θ extremely high value negatively affected the MPC but, the GP as able to learn properly and track the trajectory well. An explanation of this behaviour could be because only the angular velocity error was used as the target values within the GP model, other states

didn't affect the system as much. Even though the MPC seems to diverge, the angular velocity could be sufficiently explored, leading to a reasonably trained GP model.

κ_θ	MPC RMSE(m)	GPMPC RMSE(m)
0.1	0.3241426	2.8508358
0.5	0.0881031	2.4240400
0.6	0.0663602	3.9602015
1.0	0.0393253	1.23138746
1.6	0.0346110	0.0270105
1.8	0.0336494	0.0242535
4.0	0.0304251	0.0225876
5.5	0.0302255	0.0230424
5.6	0.1843515	0.0231803
50	166.82727	0.0543909
10000	9.2431265	0.03571508
20000	8.97530775	0.04331383

FIGURE 17 MPC AND GPMPC TRAJECTORY FOR $K_\theta = 0.1$ FIGURE 18 MPC AND GPMPC TRAJECTORY FOR $K_\theta = 0.5$

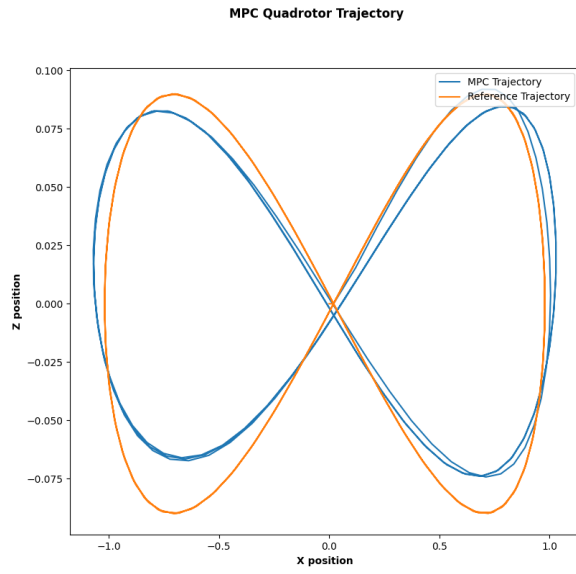


FIGURE 19 **FIGURE 18** MPC AND GPMPc TRAJECTORY FOR $K_\theta = 1.0$

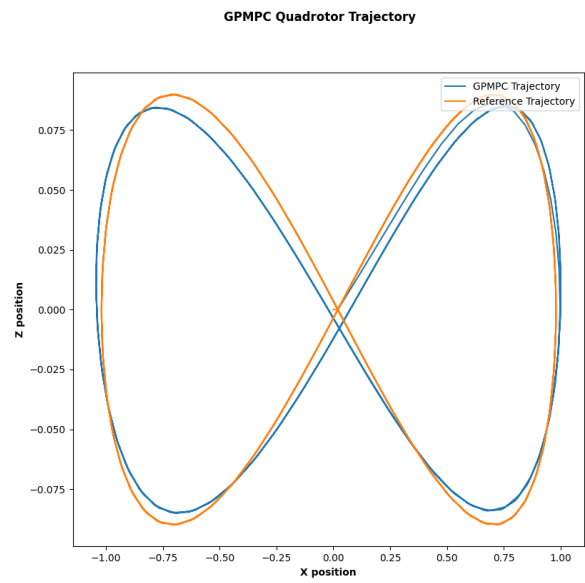
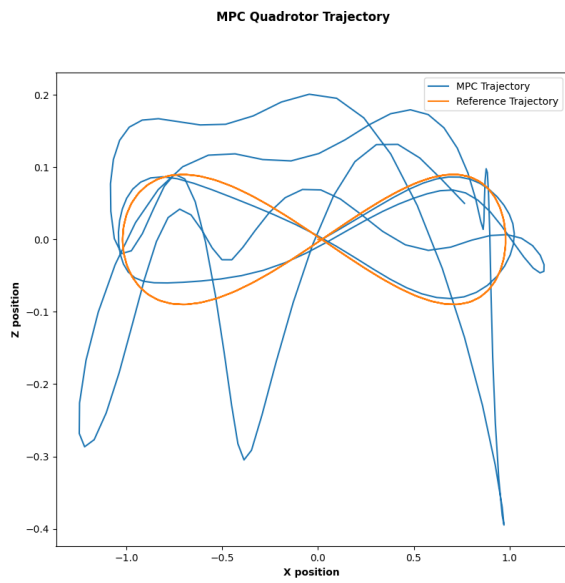


FIGURE 20 MPC AND GPMPc TRAJECTORY FOR $K_\theta = 5.6$

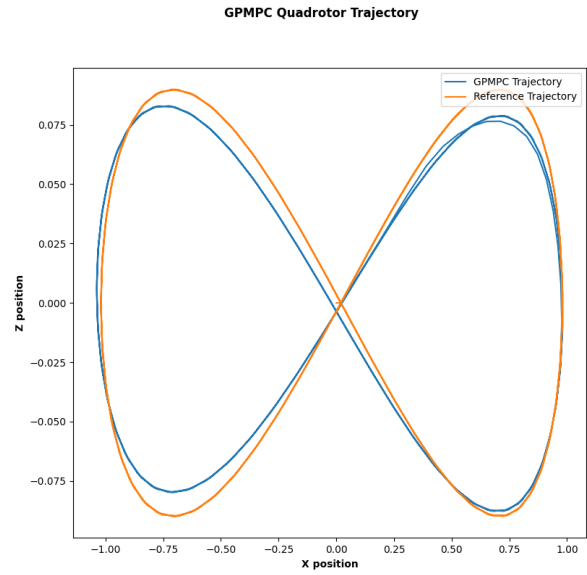
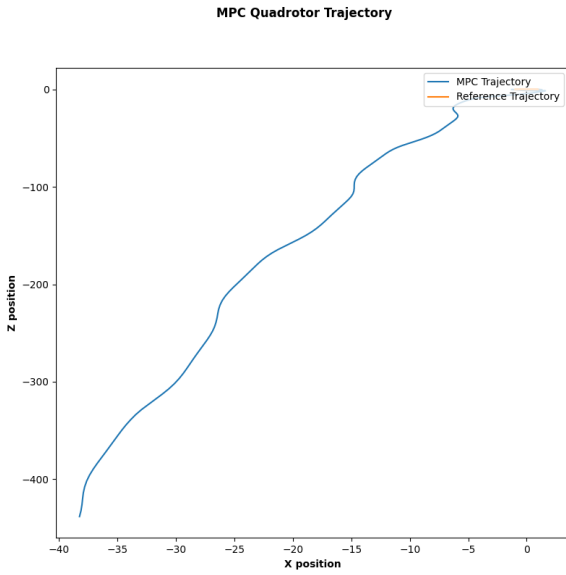


FIGURE 21 MPC AND GPMPc TRAJECTORY FOR $\kappa_\theta = 50$

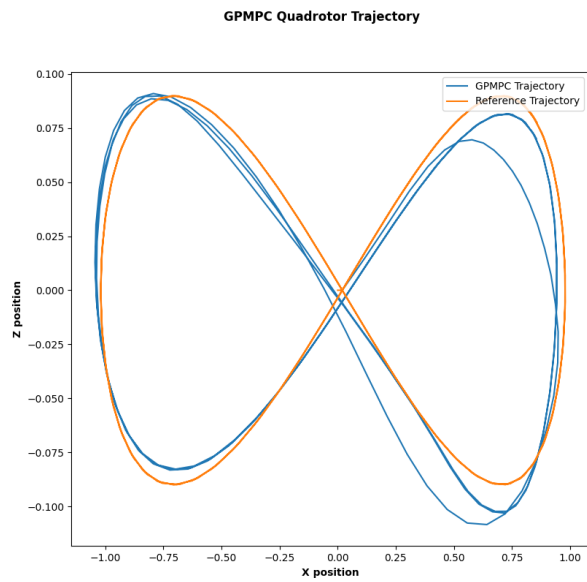
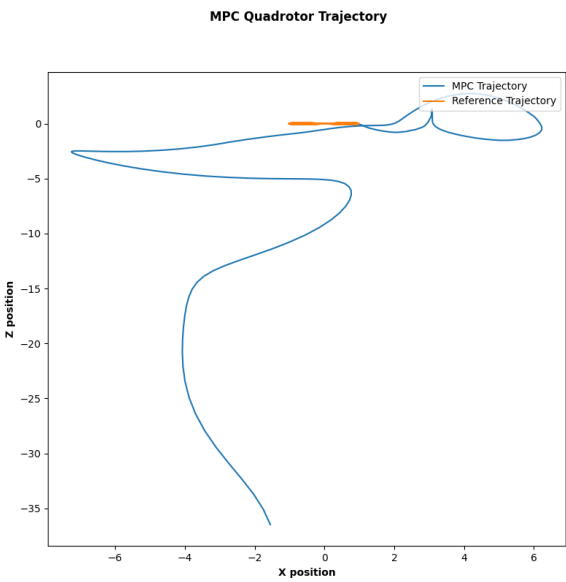


FIGURE 22 MPC AND GPMPc TRAJECTORY FOR $\kappa_\theta = 10000$

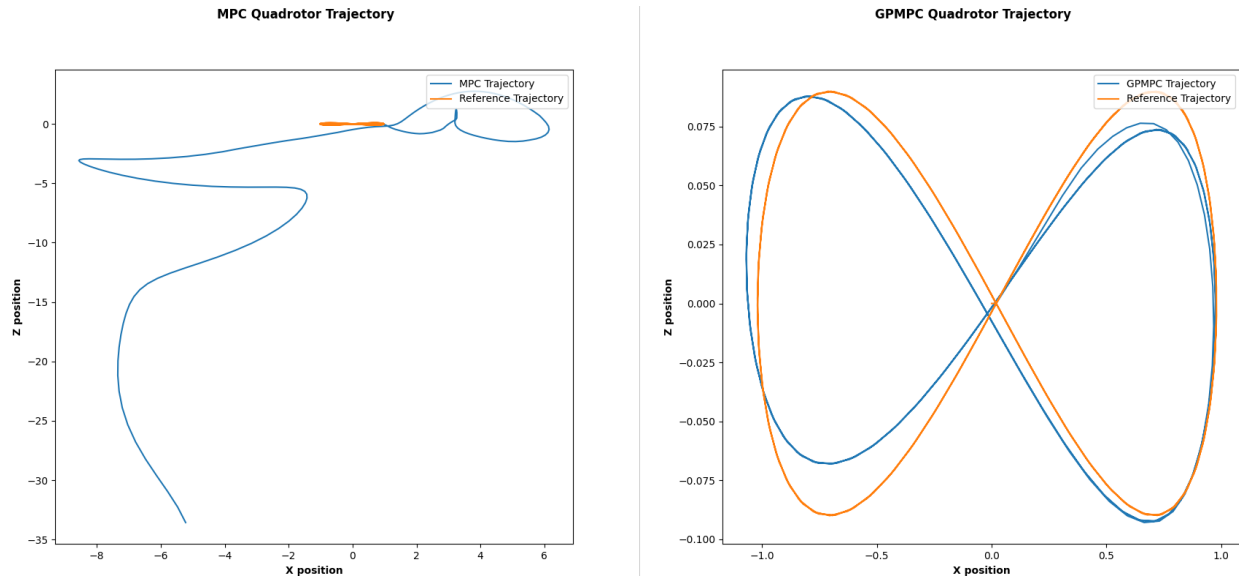


FIGURE 23 MPC AND GPMPC TRAJECTORY FOR $K_{\theta}=20000$

Radial Basis Function and Matern Kernel

In this section of the experiment, the RBF and Matern functions were utilized and compared on their performance on the lemniscate trajectory. As seen in the table below, the RBF kernel shows a better result. RBF kernel has a simple form and is more computationally efficient, a Matern kernel is more data dependent. The Matern kernel does not have “concentration of measure problem for high dimensional inputs to the extent of the RBF (Gaussian) kernel”.

Kernel	GPMPC RMSE(m)
Radial Basis Function	0.0281328
Matern	0.0301938

Noisy and Noiseless Training Data

The table below shows the lengthscale for each of the states and control inputs for the oval trajectory on the RBF kernel. A small lengthscale implies that nearby points in the input space

will have a strong influence on each other, and a larger lengthscale results in smoother functions that capture longer-range variations.

States and Input	Lengthscale With added Noise	Lengthscale With No Noise
X Position	8.6417	1.6514
Z Position	8.3996	6.1410
X Velocity	8.4174	5.1415
Z Velocity	8.7732	4.0281
Theta	8.0363	1.2817
Angular Velocity	8.4278	2.7297
Theta Input Control	7.3619	0.2195
Thrust Input Control	5.9449	5.6717

TABLE 4 LENGTHSCALE OF TRAINED DATA WITH AND WITHOUT NOISE.

The states at each second are plotted in figure 25,26, and 27 to visualize the performance of how well the MPC and GPMPC behave in comparison to the reference. The states of the GPMPC and thrust with added noise closely match that of the MPC with minor errors. The θ control for both GPMPCs started smoother but didn't track the exact path. In the plots shown in 30 and 31, the performance of the GPMPC without the added Gaussian noise was poor. As seen in figure 27, the value of θ changed more rapidly, greatly affecting the velocity and position of the quadrotor.

MPC

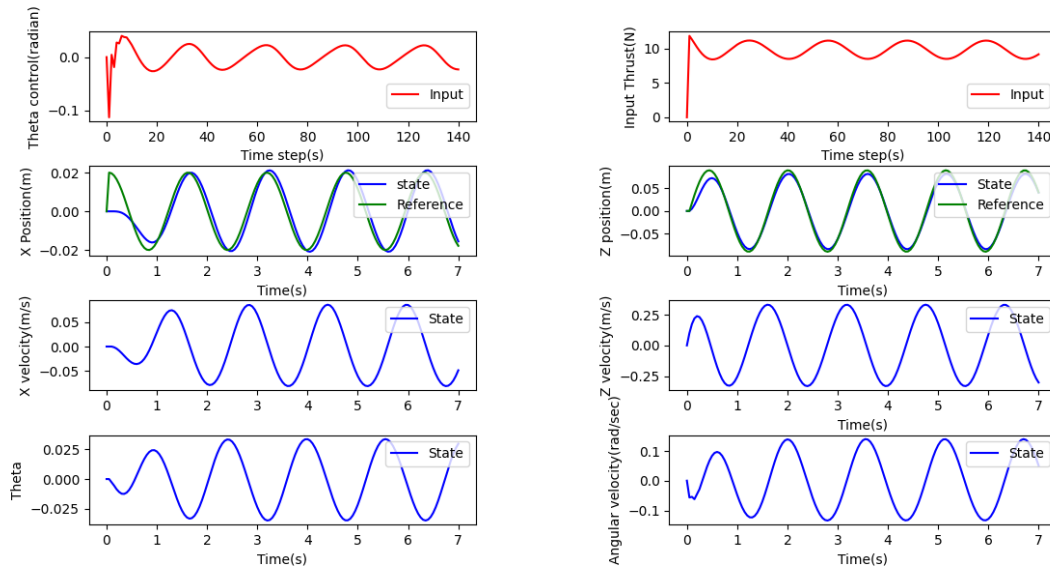


FIGURE 24. MPC + UNMODELED ERROR.

GPMPC

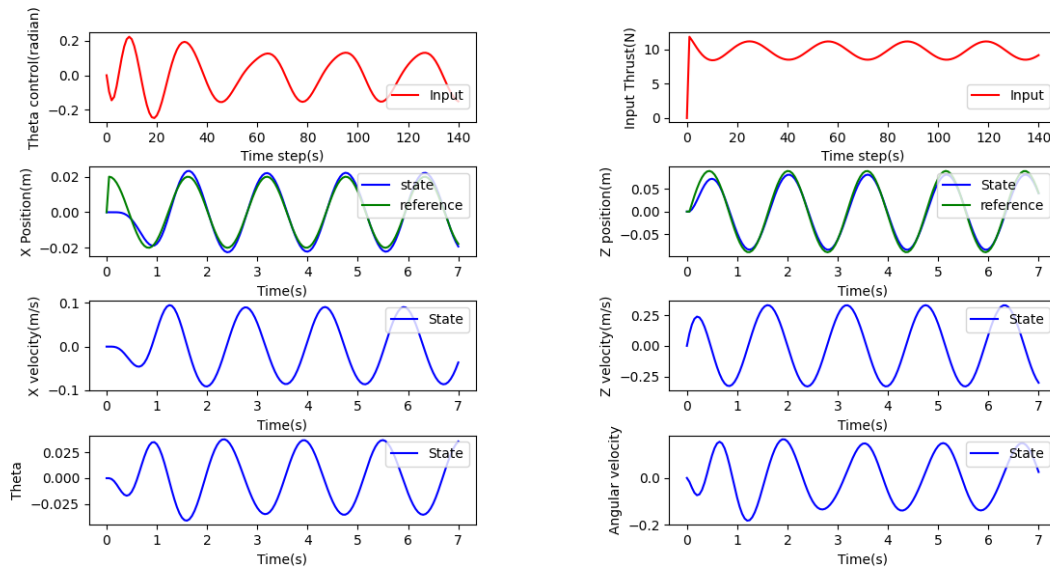


FIGURE 25. GPMPC WITH NOISE.

GPMPC

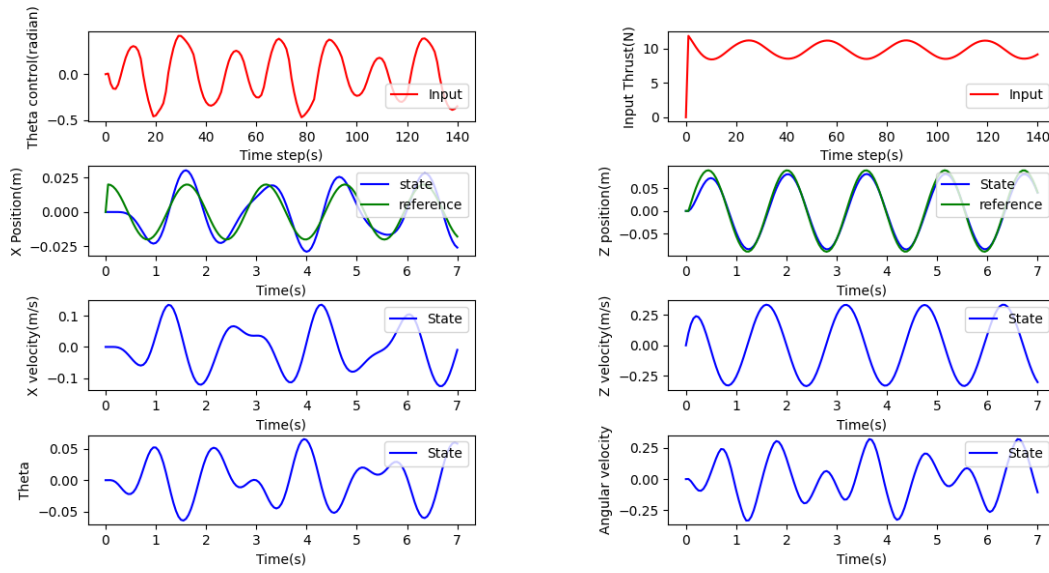


FIGURE 26. GPMPC WITHOUT NOISE.

The Error bars in the gaussian prediction were shown to represent the upper and lower bound confidence region. The bars shown in the GP with added noise is lower due to having a reduced uncertainty. From the plot of its trajectory, it's clear that it didn't create a reliable error model. The trained points on the GP prediction without added noise were further away from the training data when compared to the on with noise.

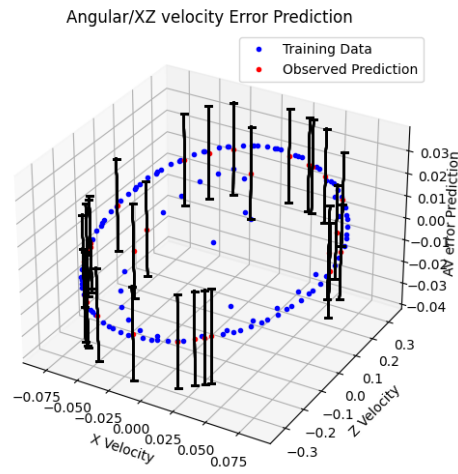


FIGURE 27. GAUSSIAN PREDICTION PLOT WITH NOISE.

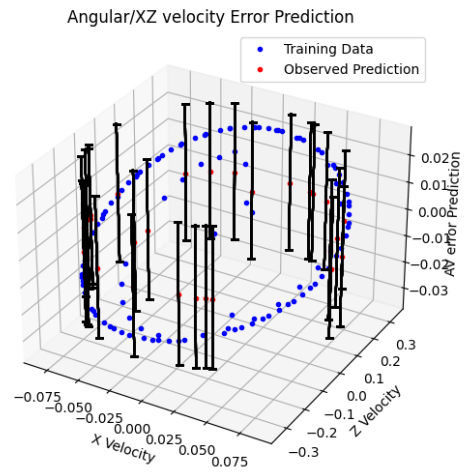


FIGURE 28. GAUSSIAN PREDICTION PLOT WITHOUT NOISE.

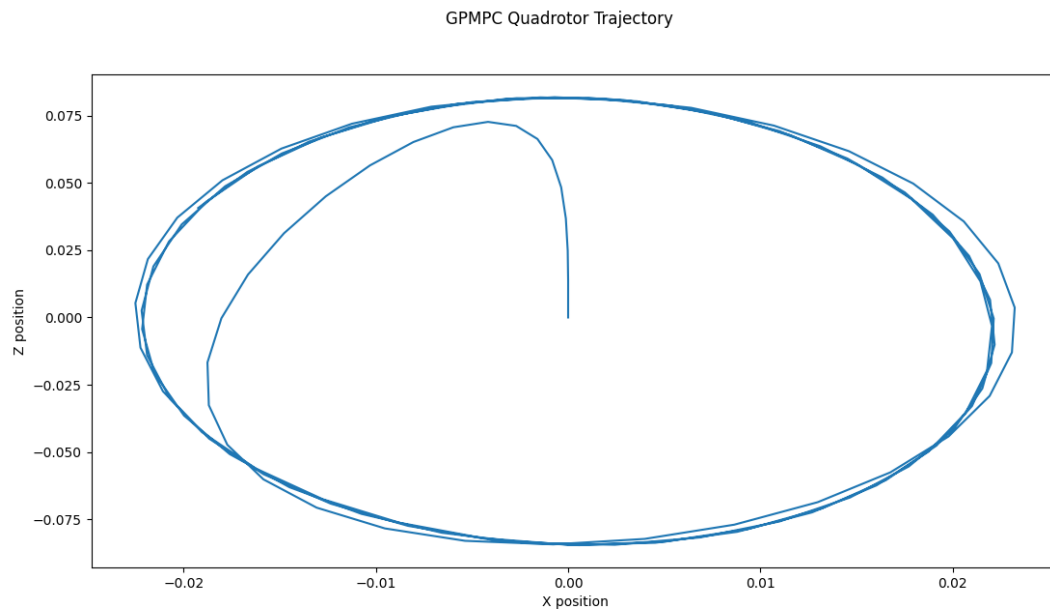


FIGURE 29. GMPC OVAL TRAJECTORY WITH ADDED TRAINING NOISE.

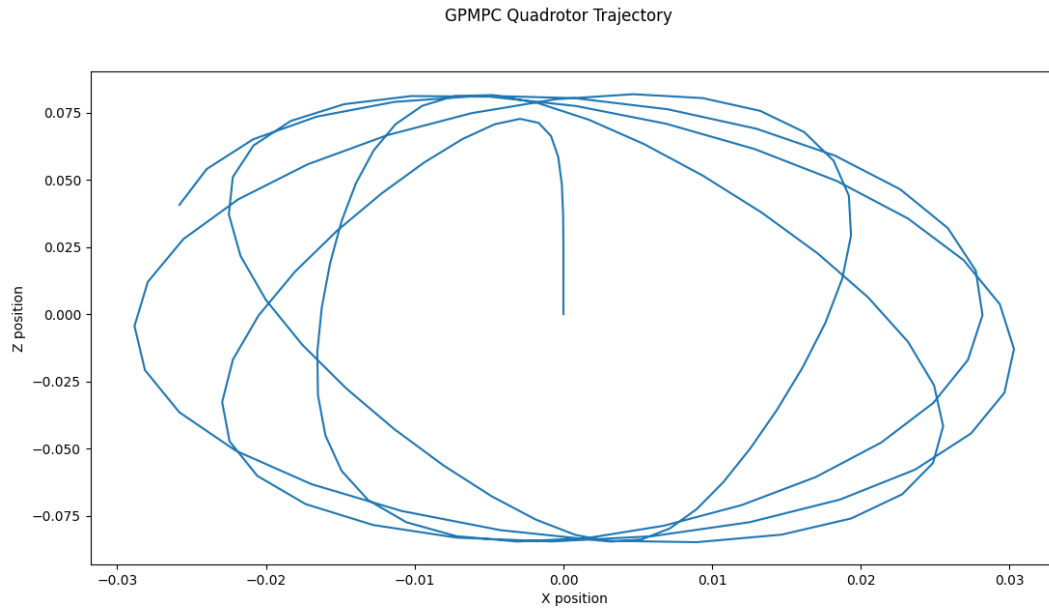


FIGURE 30. GPMPC OVAL TRAJECTORY WITHOUT ADDED TRAINING NOISE.

Conclusion

In the scope of a safe learning-based control strategy, the GPMPC has proven to show a promising path for advanced UAV navigation. The primary objective of this project was to display the potential of GPMPC. By combining the power of GP learning with MPC, drones can be designed not only to respond to commands but to adapt and improve their flight performance over time. In the result section, the outcomes from changing different variables inside the GPMPC provided a good understanding on the benefits and limitation of GP. The learning-based approach using GP allows for a more reliable trajectory tracking due to its ability to compensate for modelling errors and its flexibility to learn with a relatively small data set. Based on the tests performed, drawbacks such as the scaling computational intensity needed for large data sets and the tuning of control parameters in the MPC for the GP to learn properly, could result in a subpar performing GPMPC when compared to MPC. The issues of overfitting and the choice of kernel are also reasons why a GP would not improve an MPC's performance.

Limitation

Although the result was satisfactory, some additional test could be added to study the GPMPC better. Improvements such as running the GP multiple times and learning from the previous error difference should ideally improve the performance each run. Increasing the simulation time for the quadrotor and moving it through a more complex trajectory, to make the system sufficiently exciting. By doing this the system captures a diverse range of scenarios to improve its predictivity and performance but adding more data increases the computational expense. By utilizing more target states in the GP model, not only angular velocity error, this would allow the GP to learn more and make better predictions. The GP was trained on specific trajectories, it could be improved by training on a wider range of states, allowing the same GP to be used on any trajectory.

References

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- [2] A. Girard and R. Murray-Smith, “Gaussian processes: Prediction at a noisy input and application to iterative multiple-step ahead forecasting of time-series,” *Switching and Learning in Feedback Systems*, pp. 158–184, 2005. doi:10.1007/978-3-540-30560-6_7
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